

● TEAM ENSIUMBLE

Machine Learning Project

25/26

Model comparison, ensembling and a
stacking meta-learner across the MONK
benchmarks and ML_CUP

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Project type B · scikit-learn · PyTorch

Headline results

CLASSIFICATION

100%

Test accuracy on MONK-1
and MONK-2

NOISE CEILING

97.22%

MONK-3, reached by four
model families alike

REGRESSION

14.3289

Final ML_CUP test MEE via
RidgeCV stacking

IMPROVEMENT

−45.3%

Against the OLS baseline of
26.1816

Six model families — linear, regularised linear, K-NN, kernel SVM and PyTorch MLPs — run twice:
on synthetic MONK classification, and on high-dimensional ML_CUP regression.

Scope and experimental protocol

MONK 1-3 · CLASSIFICATION

Diagnostic, not competitive. Each family is tuned rigorously, but no cross-family selection is made — the deliverable is an argument about hypothesis space and data quality.

6 categorical features → 17 binary · 432 test samples

ML_CUP · REGRESSION

Error is Mean Euclidean Error over a four-dimensional target:

$$\text{MEE} = \frac{1}{N} \sum_{i=1}^N \|\hat{y}_i - y_i\|_2$$

5-fold CV · seed 42

Every hyperparameter search is scored identically and reproducibly.

400 / 100 split

80% carries all model selection; 20% stays untouched until final assessment.

Glorot / He init

Activation-specific schemes; classification nets use BCE with a sigmoid output.

01

MONK's classification benchmark

Three problems, one controlled ladder — isolating a hypothesis space that is too poor from a target that is partly unlearnable.

MONK-1 — non-linear, but only just

SOLVED · 100.00%

MODEL	SELECTED HYPERPARAMETERS	TEST ACC.	MSE
Logistic Regression	L1 · C = 0.1 · saga	75.00%	0.2500
Ridge Classifier	L2 · λ = 0.001	73.61%	0.2639
K-NN	K = 15 · uniform · p = 1	77.31%	0.2269
SVM	poly · degree 2 · C = 10 · coef0 = 1	100.00%	0.0000
PyTorch MLP	1 hidden layer · 2-4 units · BCE	100.00%	0.0000

Logistic Regression records zero false positives alongside 108 false negatives — the signature of a boundary that cannot bend. Degree 2 winning over degree 3 says the required interactions are simple.

MONK-2 — linear models collapse into the class prior

SOLVED · 100.00%

Logistic Regression $L_2 \cdot C = 0.01$ 67.13%

Ridge Classifier $L_2 \cdot \lambda = 500$ 67.13%

K-NN $K = 15 \cdot \text{distance} \cdot p = 2$ 75.93%

SVM $\text{poly} \cdot \text{degree } 3 \cdot C = 10$ 85.65%

PyTorch MLP $1 \text{ layer} \cdot 2 \text{ units} \cdot \text{ReLU} \cdot \text{Adam}$ 100.00%

Logistic Regression confusion matrix

432 test samples · AUC 0.56

	PRED 0	PRED 1
TRUE 0	290	0
TRUE 1	142	0

Every prediction falls in the left column: 67.13% is the class prior, not a classifier.

Polynomial and RBF maps are fixed in advance; the network learns an intermediate representation fitted to this target. When the required feature transform is unknown, learned features beat designed ones.

MONK-3 — every family stops at the same number

CEILING · 97.22%

Logistic

97.22%

L1 · C = 1

Ridge

97.22%

L2 · $\lambda = 10.0$

K-NN

89.12%

K = 11 · uniform

SVM

97.22%

linear · C = 0.1

PyTorch MLP

97.22%

3 hidden layers

ENGINEERING TAKEAWAY

K-NN is the one model that memorises individual training points, and the one model that degrades. Mislabeled samples inside a K = 11 neighbourhood corrupt the vote, while parametric models fit a global boundary and average the corruption away. The binding constraint is data quality, not model capacity — further architecture search here is wasted budget.

02

ML_CUP regression and dimensionality

A collinear 12-feature input space, a four-dimensional target, and a baseline to beat.

Linear baselines and regularisation

BASELINE · OLS

26.1816

Unregularised least squares. Everything after this is measured against it.

L1 · LASSO

25.6353

Winner $\alpha = 0.1$ over a six-point log grid from $1e-4$ to 10.

L2 · RIDGE

25.5736

Winner $\alpha = 100$ — a heavy penalty winning is itself the diagnosis.

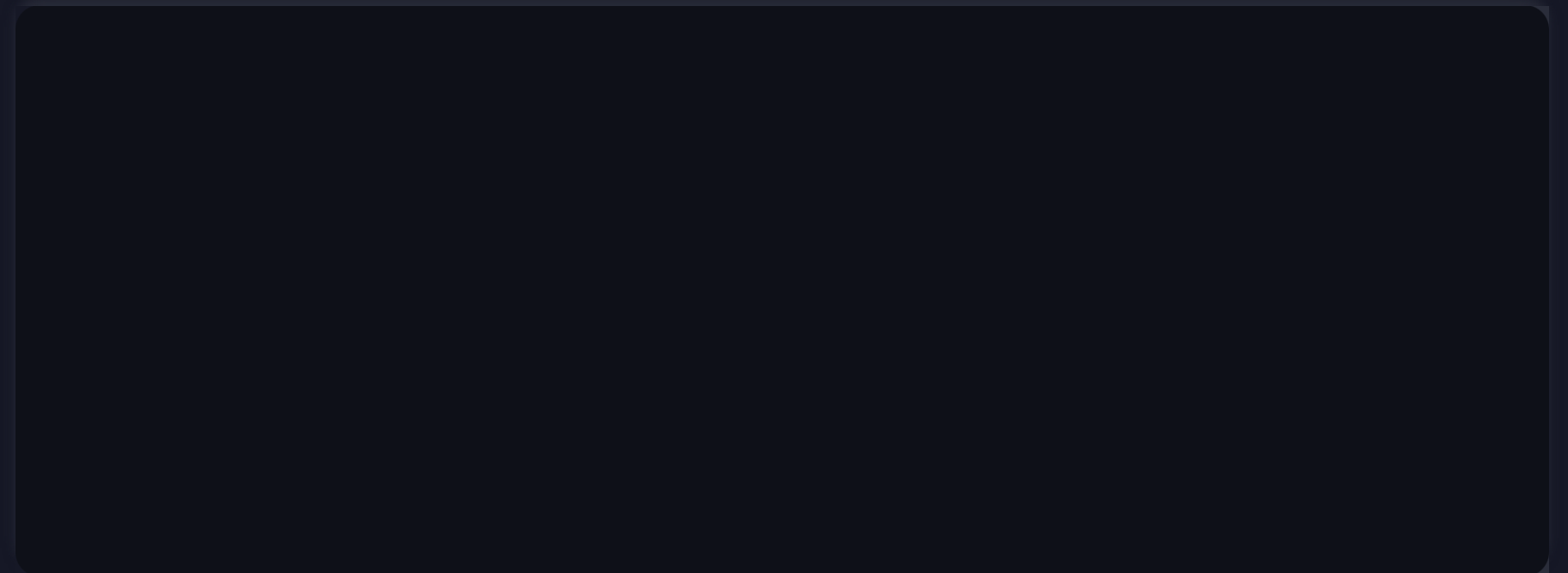
Both regularisers improve on plain least squares, which points at either noise or collinearity. PCA settles which.

Principal component analysis

> 95%

of the variance in the 12-dimensional input is explained by the first principal component alone. The features are near-collinear — which is exactly why a strong ridge penalty helped.

The caveat is deliberate: high input variance does not imply high correlation with the target. So the number of components entered the grid as a hyperparameter over [1, 12], mostly to give K-NN a fighting chance against the curse of dimensionality.



K-NN and SVR model selection

K-NN

15.1070

PCA = 6 · K = 3 · weights = distance

Searched over $K \in \{3, 4, 5, 7, 10, 15, 20, 30\} \times \{\text{uniform, distance}\} \times \text{PCA} \in [1, 12]$. A small K winning on six components is the mirror image of MONK: the local neighbourhood is informative once redundant dimensions are stripped out.

SVR

15.5959

PCA = 4 · RBF · C = 3 · γ = 3

All four kernels from the MONK study were re-run here — linear, RBF, polynomial (degree 1–4) and the two-layer perceptron kernel. RBF wins on four components with a moderate C: a smooth fit rather than a tightly-fitted one.

MLP grid search and optimisation

FIRST SEARCH · WINNER C0169

arch = [32, 16] · act = tanh · opt = adam lr =
0.005 · wd = 0.001 · dropout = 0 · bs = 32

21.952 ± 0.844

FINAL SEARCH · WINNER C0320

arch = [256, 256, 128] · act = gelu · opt =
adamw lr = 0.02 · wd = 0.1 · dropout = 0.1 · bs
= 32

18.058 ± 1.168

batch = 32

Small batches consistently
minimised MEE — more
gradient noise, better
generalisation.

GeLU

By far the best of five
activations (Tanh, ReLU,
GeLU, SeLU, eLU).

Cosine annealing

A general improvement over
ReduceLROnPlateau across
configurations.

Adam / AdamW

The two best optimisers —
and they disagree on weight
decay, which stage 2 exploits.

The generalisation gap that tuning would not close



C0320 learning curves. Validation flattens near 18.6 while training keeps falling past 11 — a gap of more than five MEE points that no further hyperparameter tuning closed.

The response was to stop optimising bias and start attacking variance.

03

Ensembling and the stacking meta-learner

Three stages of increasing heterogeneity: same model different seeds, then different models, then different families.

The ambiguity decomposition

$$E_{\text{ensemble}} = \bar{E} - \bar{A}$$

Ensemble error equals the average member error minus the ambiguity — the variance among member predictions. Raising heterogeneity raises ambiguity, and is the only lever that lowers the ensemble's error without making any single member better.

Stage 1 · seeds

Stage 2 · optimisers

Stage 3 · model families

STAGE 1

Multi-seed ensemble — one configuration, five seeds

SEED	TRAINING MEE	VALIDATION MEE
Seed 42	12.7416 ± 3.6794	18.0583 ± 1.1681
Seed 123	15.7754 ± 5.0723	19.5028 ± 1.9016
Seed 456	17.6701 ± 3.4485	20.5135 ± 1.0528
Seed 789	12.5222 ± 4.0270	18.5787 ± 1.1755
Seed 2024	13.7924 ± 5.0265	19.0560 ± 2.2041
Ensemble	13.5540 ± 2.1089	18.2175 ± 0.7553

The final grid-search winner — [256, 256, 128], AdamW, lr 0.02, dropout 0.1, batch 32 — retrained under five seeds and averaged.

18.2175 vs 19.1419 member mean
gap 5.3168 → 4.6635
 σ 1.1681 → 0.7553

Seed 42 turns out to have been lucky. Averaging buys stability, not a better headline number.

STAGE 2 Ten heterogeneous MLPs — Adam against AdamW

Adam's best runs prefer weight decay near 0.001; AdamW's prefer 0.05–0.1. Two regularisation regimes reaching comparable validation error is free ambiguity — so the ensemble took the best five of each.

TRAINING MEE

11.6355

VALIDATION MEE

17.390

GAP

5.7545

The trade is explicit: the gap did not shrink, but validation MEE broke below 18 for the first time. Heterogeneity across regularisation regimes buys accuracy; stage 1 bought stability. Neither buys both.

STAGE 3

Stacking meta-learner architecture

```
BASE LEARNER · MLP  
[256, 256, 256] · adamw · gelu · 68 epochs
```

```
BASE LEARNER · K-NN  
K = 3 · weights = distance · PCA = 6
```

```
BASE LEARNER · SVR  
RBF · C = 3 · γ = 3 · PCA = 4
```



out-of-fold
predictions

```
META-LEARNER · RIDGECV, A = 10.0
```

$$w^* = \arg \min_w \|Zw - y\|_2^2 + \alpha \|w\|_2^2$$

Generalised cross-validation selects $\alpha = 10.0$. The curve is flat below $\alpha \approx 100$ and collapses above 10^3 , so the solution sits comfortably inside a stable region rather than balanced on an optimum.

Because the earlier winner was seed-dependent, the MLP member was chosen by re-running all ten configurations across five seeds and ranking on the mean, not the best.

Final assessment on held-out data

BASE LEARNER	TR MEE	VL MEE (OUT-OF-FOLD)	TS MEE
MLP	11.8339	18.9522	17.8870
K-NN	0.0000	15.1070	14.1284
SVR	8.0918	15.5959	15.7672
Stacking · RidgeCV $\alpha = 10.0$	5.9037	14.3715	14.3289

K-NN's zero training error is memorisation, not skill — which is exactly why it is useful in the stack: its error pattern is uncorrelated with the MLP's. The meta-learner beats every member on out-of-fold validation and matches the best member on test, with the lowest validation-to-test drift of any model in the study (0.0426) — it buys stability across partitions rather than a lower headline number.

Error trajectory across the study



Engineering takeaways

CAPACITY IS NOT ALWAYS THE CONSTRAINT

MONK-3 caps four different model families at the same 97.22%. When capacity stops mattering, the bottleneck has moved to the data — and more architecture search is wasted budget.

LEARNED FEATURES BEAT DESIGNED ONES

A two-unit network solves MONK-2 where a degree-3 kernel reaches 85.65%. Fixed feature maps lose to adaptive representations when the required transform is unknown.

EACH STAGE MOVED A DIFFERENT QUANTITY

Averaging seeds cut variance without improving the headline; averaging optimisers improved the headline without closing the gap; stacking across families did both, because the meta-learner weights members by where they are reliable.

THE TRADE-OFF IS REAL

The final model costs three pipelines plus a meta-fit for a 45.3% improvement over OLS baseline. Worth it when scored on generalisation; for a latency-bound deployment, K-NN alone at 14.1284 is the honest first choice.